

Predictive Risk Analytics in Retail Banking: A Comparative Study of Machine Learning Models for Financial Fraud Detection

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Abstract—Financial fraud has become a major challenge in the banking industry due to the increasing adoption of digital transactions and the complexity of fraudulent behaviours. Traditional rule-based fraud detection approaches may struggle to identify evolving transaction patterns, creating a need for data-driven machine learning solutions. This study aims to develop and compare machine learning classification models for detecting fraudulent banking transactions using a retail banking transaction dataset. The study evaluates three classification algorithms, namely Logistic Regression, Decision Tree, and Random Forest, based on their ability to distinguish fraudulent and legitimate transactions. The dataset consists of 10,000 transaction records with a moderate class imbalance, where fraudulent transactions represent approximately 12.5% of the total data. Data preprocessing involved duplicate removal, irrelevant attribute removal, numerical feature scaling, categorical feature encoding, and cost-sensitive learning using class weighting. In addition, two behavioural features, namely Login Attempt Ratio and Transaction Amount per Account Age Ratio, were engineered to capture suspicious transaction patterns. Exploratory data analysis and Mutual Information analysis were performed to examine feature relationships and identify influential attributes related to fraud prediction. Model performance was evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and 5-fold cross-validation. The results showed that Random Forest achieved the highest accuracy and precision, while Logistic Regression demonstrated superior fraud detection capability with the highest recall and ROC-AUC score. Therefore, Logistic Regression was selected as the best baseline model and optimized using GridSearchCV with 5-fold cross-validation. The tuned Logistic Regression model achieved a testing ROC-AUC score of 0.9797 and recall of 0.96, demonstrating strong capability in identifying fraudulent transactions while maintaining good generalization performance. The findings indicate that Logistic Regression provides a reliable and effective approach for retail banking fraud detection.

Keywords—*Fraud detection, machine learning, logistic regression, random forest, retail banking*

I. INTRODUCTION

Digital convenience encourages the growth of digital banking services and online financial transactions. This evolution, however, also introduces new challenges, increasing the complexity and volume of financial data while complicating

the accurate and efficient decoding of customer behavior [1]. According to Clark (2021), as cited in [2], the rise of online banking has led to a significant increase in identity theft. Banking fraud results in massive financial losses and erodes customer trust. For example, official records show that the domestic financial impact of reported digital fraud and scams reached RM 2.8 billion in 2025 [3]. Developing an accurate fraud detection system is therefore a vital area in financial risk management [2].

Previous research has explored rule-based methods for fraud detection, but the dynamic nature of financial fraud requires the continuous exploration of new approaches [2]. To address the flaws of static rules, financial institutions utilize machine learning. Machine learning is an analytical approach where computational algorithms study historical data to identify complex trends and make automated predictions without explicit programming [4].

Fraud detection is a challenging machine learning problem because fraudulent transactions represent a very small proportion of overall transaction volume. Furthermore, Huang (2019), as cited in [2] stated that transaction behaviors vary among users; thus, applying a uniform threshold to detect anomalies often leads to higher false-positive rates for certain individuals. For this reason, evaluation metrics such as recall, F1-score, and ROC-AUC are critical alongside standard accuracy.

This study focuses on developing a predictive risk analytics approach for retail banking fraud detection by comparing three machine learning classification models: Logistic Regression, Decision Tree, and Random Forest. The objectives of this study are (1) To analyse transaction data and identify important features related to fraudulent activities, (2) To compare the performance of different machine learning classification models for fraud detection, and (3) To optimize the selected model using hyperparameter tuning and evaluate its capability in detecting fraudulent transactions.

II. LITERATURE REVIEW

Machine learning has been widely applied in financial fraud detection due to its ability to analyze large transaction datasets and identify complex patterns [1]. Previous studies have extensively demonstrated that standard classification

algorithms, specifically Logistic Regression [5], Decision Trees [1][6], and Random Forests [7], can be effectively utilized to distinguish between fraudulent and legitimate transactions.

Logistic Regression is one of the most used classification algorithms due to its simplicity, interpretability, and ability to estimate the probability of an event [5]. In fraud detection, Logistic Regression provides valuable insights into how different transaction characteristics influence the likelihood of a fraudulent occurrence [2]. Although it assumes a linear relationship between features and outcomes, it often performs exceptionally well when combined with proper data preprocessing and robust feature selection techniques [7].

Decision Tree models are frequently applied in fraud detection because they can capture non-linear relationships and provide easily interpretable decision rules [6]. However, individual Decision Tree models are prone to becoming overly complex and learning noise from the training data, which ultimately results in overfitting. Therefore, controlling tree depth and regulating other hyperparameters is critical to improve the model's generalization to unseen data [2].

Random Forest is widely utilized as an ensemble learning method that combines multiple Decision Trees to enhance overall prediction performance [7]. By aggregating multiple trees, Random Forest effectively reduces model variance and handles the highly complex patterns inherent in financial transaction data [1]. While previous research has consistently shown that ensemble methods achieve strong predictive accuracy in fraud detection tasks, these models must still be carefully optimized, as high model complexity can still lead to overfitting [5].

Beyond model selection, feature engineering and feature importance analysis are vital components in optimizing overall fraud detection performance [5]. Transaction-related features, such as unusual transaction amounts, login behaviors, authentication information, and suspicious activity patterns, provide critical signals for isolating fraudulent behavior [2]. Within this framework, Mutual Information is a useful feature importance analysis approach because it quantifies the dependency between individual input features and the target variable [8].

Due to the inherently imbalanced nature of financial fraud datasets, relying solely on classification accuracy often yields

highly misleading conclusions [2]. Therefore, performance metrics such as recall, F1-score, and ROC-AUC are prioritized, as they provide a significantly more comprehensive and realistic evaluation of a model's true capability to isolate rare fraudulent cases [5].

III. RESEARCH METHODOLOGY

This section outlines the structured workflow utilized to build and assess the financial fraud detection models. The methodology follows a systematic sequence of steps: data preparation, data preprocessing, exploratory descriptive analysis, model development, and final evaluation. Figure 1 illustrates the comprehensive layout of this research process.

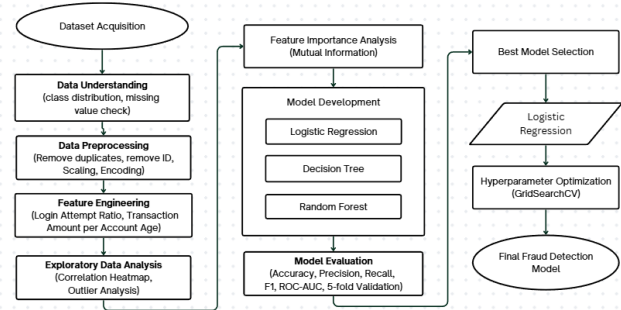


Figure 1. Research Framework

A. Data Preparation

The dataset used in this study is a retail banking transaction dataset containing 10,000 transaction records and 20 attributes [9]. The dataset includes transaction information, user behaviour indicators, authentication details, and the target variable fraud_flag. The target variable represents whether a transaction is fraudulent (True) or legitimate (False).

The dataset contains 17 numerical features and 2 categorical features. The numerical variables include ratio and interval metrics such as Transaction_amount, Login_attempts, Device risk score, and Transfer frequency. The categorical variables are Payment channel and Authentication type. The final attribute is the binary target variable, fraud_flag. This target indicates whether a specific transaction is verified as fraudulent (True) or legitimate (False). Table 1 details the characteristics and measurement levels of these features.

TABLE I. DESCRIPTION OF DATASET ATTRIBUTES

Attribute	Type	Level	Description
Transaction_id	Cat.	Nominal	Unique cryptographic or numerical identifier assigned to each individual transaction record.
Transaction_amount	Num.	Ratio	The absolute monetary value of the financial transaction being processed.
Login_attempts	Num.	Ratio	The total number of authentication attempts recorded during the user session.
Device risk score	Num.	Interval	A calculated safety metric evaluating the risk profile of the hardware/device used to initiate the transaction.
Transfer frequency	Num.	Ratio	The historical frequency or cumulative count of fund transfers executed by the account holder.
Anomaly frequency	Num.	Interval	A predictive baseline score tracks how drastically current transaction parameters deviate from historical user behavior profiles.
Account age days	Num.	Ratio	The total chronological lifespan of the bank account, measured in days from registration to the transaction date.
Transaction time hour	Num.	Interval	The exact hour of the day (scaled from 0 to 23) when the transaction request was submitted.

Attribute	Type	Level	Description
Failed transactions last 30 days	Num.	Ratio	The rolling total of unsuccessful or declined transaction events recorded on account over the previous 30 days.
Average monthly balance	Num.	Ratio	The historical mean financial balance maintained by the customer across monthly cycles.
Daily transaction count	Num.	Ratio	The total volume of distinct transaction requests completed within the current 24-hour window.
Geo distance (km)	Num.	Ratio	The calculated physical distance in kilometers between the transaction point of origin and the cardholder's primary home location.
Session duration (min)	Num.	Ratio	The total elapsed active time of the user's online or mobile banking session, measured in minutes.
Transaction velocity score	Num.	Interval	A real-time monitoring metric tracking the speed and frequency of successive transactions over a highly compact time frame.
Payment channel	Cat.	Nominal	The technological medium used to route the payment (e.g., Web Banking, Mobile Application, Point of Sale, ATM).
Authentication type	Cat.	Nominal	The primary verification mechanism utilized to authorize the transaction (e.g., Password, PIN, SMS One-Time Password, Biometrics).
Card present flag	Num.	Nominal	Binary indicator mapping whether the physical credit/debit card was present at the vendor terminal (1: Yes, 0: No).
International transaction flag	Num.	Nominal	Binary indicator specifying if the transaction was cleared across international sovereign borders (1: Yes, 0: No).
Suspicious ip flag	Num.	Nominal	Binary indicator flagging whether the originating network IP address matches known blacklists or high-risk geolocation pools (1: Yes, 0: No).
Fraud_flag	Bin.	Nominal	0: No, 1: Yes. The target classification label indicates whether the transaction was verified as legitimate or fraudulent.

B. Data Preprocessing

Prior to model training, a series of data preprocessing operations were executed to optimize data quality and maximize classifier stability [7]. First, duplicate records were isolated and removed to prevent recurrent transactional observations from biasing the machine learning process. Next, the dataset was evaluated for missing values and anomalies. There were no missing data points across the attributes. Outliers in transaction amounts were kept in the dataset rather than removed, as extreme values are a common characteristic of genuine fraudulent behavior.

Following data cleaning, numerical and categorical attributes were processed using separate pipelines. Numerical features were standardized using a Z-score method via StandardScaler. This step transforms continuous variables to share a mean of zero and a standard deviation of one. Feature scaling is necessary to improve gradient descent optimization and regularization performance in linear models like Logistic Regression [7]. Categorical features, specifically payment_channel and authentication_type, were processed using One-Hot Encoding. This process converts discrete string variables into binary vectors so that mathematical classification models can read them.

To handle the highly imbalanced class distribution without losing critical data through undersampling or creating artificial variance through oversampling, cost-sensitive learning was applied. This method uses algorithm-level class weighting by setting the parameter class_weight='balanced' [10]. This technique adjusts the loss function during model training by penalizing mistakes on the minority fraud class more heavily [10]. Therefore, the models are forced to prioritize learning the hidden patterns of rare fraudulent transactions [10].

C. Feature Engineering

Raw transactional data frequently lacks the behavioral context necessary to expose sophisticated fraud strategies.

Therefore, feature engineering was performed to construct domain-specific behavioral indicators designed to enhance the classification boundaries of the predictive models [11]. Two novel interaction attributes were engineered: Login Attempt Ratio and Transaction Amount per Account Age.

1) *Login Attempt Ratio*. This feature quantifies the operational frequency of user authentication actions relative to the chronological lifespan of the account. A disproportionately high ratio serves as a strong proxy for suspicious or anomalous access behavior, an occurrence frequently observed when compromised or newly generated accounts undergo credential testing. Mathematically, the metric is formulated as:

$$\text{Login Attempt Ratio} = \frac{\text{Login Attempts}}{(\text{Account Age Days}) + 1} \quad (1)$$

Where *Login Attempts* represents the cumulative count of login attempts recorded, and *Account Age Days* denotes the total number of days the account has been active. The addition of 1 in the denominator serves as a smoothing parameter to prevent division-by-zero errors for newly registered accounts on day zero.

2) *Transaction Amount per Account Age*. This feature is engineered to capture high-velocity financial exposures occurring on relatively immature accounts, isolating a common paradigm where fraudulent actors rapidly exploit recently created profiles before security systems can flag them. The ratio is calculated as follows:

$$\text{Amount Account Age Ratio} = \frac{\text{Transaction Amount}}{\text{Account Age Days} + 1} \quad (2)$$

Where *Transaction Amount* represents the raw monetary value of the transaction under evaluation, and the denominator is adjusted by 1 to maintain numerical stability across all account lifespans.

These features were added to represent behavioral patterns that may not be directly captured by the standalone, original transaction attributes [11].

D. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the characteristics of the dataset before modelling. A correlation heatmap was generated to examine relationships between numerical variables and the fraud target.

E. Feature Important Analysis

Mutual Information analysis was performed to examine the relationship between each input feature and the fraud target variable [12]. Mutual Information measures the statistical dependency between independent variables and the target class, where higher scores indicate that a feature provides more useful information for distinguishing fraudulent and legitimate transactions [12]. In this study, Mutual Information was used for feature importance assessment and interpretation rather than for removing or reducing the feature set before model training.

F. Machine Learning Modelling

Three supervised classification algorithms were developed and compared: Logistic Regression, Decision Tree, and Random Forest. Logistic Regression serves as the baseline model because of its simple structure and clear interpretability. The algorithm calculates the probability of a transaction being fraudulent based on the input features. Next, a Decision Tree model was applied because it can identify non-linear relationships and create clear decision rules from transaction details. Finally, Random Forest was chosen as an ensemble learning method. This algorithm combines multiple single decision trees to improve overall prediction stability and minimize overfitting. All three models were trained using an identical data split, allocating 70% of the dataset for training and 30% for testing. A stratified split was used to maintain the original 12.5% fraud distribution across both the training and testing sets

G. Model Evaluation

The predictive performance of each model was measured using six standard evaluation criteria. First, accuracy calculates the overall percentage of correctly classified transactions. Second, precision determines the proportion of predicted fraudulent transactions that are genuinely fraudulent. Third, recall assesses the ability of the model to detect all actual fraudulent cases in the dataset. Fourth, the F1-score provides a harmonic balance between precision and recall, which is helpful for evaluating imbalanced classes. Fifth, the Area Under the Receiver Operating Characteristic Curve (ROC-AUC) assesses how well a model separates fraudulent transactions from legitimate ones across multiple classification thresholds. Last, 5-fold cross-validation was applied to assess model consistency during evaluation.

IV. RESULTS AND DISCUSSION

A. Exploratory Data Analysis

Exploratory Data Analysis (EDA) began by checking the completeness, structure, and relationships within the dataset to set a clear baseline before modeling. A data profile scan confirmed that there were no missing values across all 10,000 records. This provided a clean dataset for model training. During this phase, the transaction_id attribute was removed from the feature set. A unique transaction identifier acts as a high-cardinality nominal index. It has no statistical variance across classes and gives no predictive power for finding systemic fraud patterns.

The distribution of the target variable, fraud_flag, was evaluated to understand the balance of the binary classification problem. The dataset contains 8,749 legitimate cases and 1,251 fraudulent records.

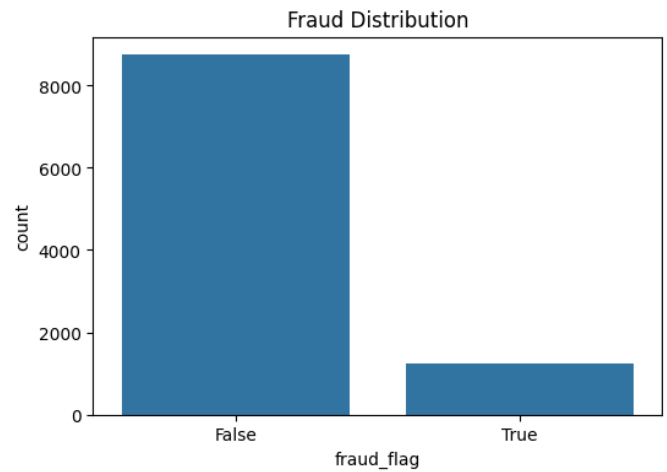


Figure 2. Target Class Distribution

As shown in Figure 2, this distribution shows a moderate class imbalance where fraudulent events make up exactly 12.5% of the total data. In fraud analytics, an imbalanced target distribution means that relying only on standard classification accuracy can create a misleading model that fails to detect actual fraud [13]. This target layout provides the reason for using cost-sensitive optimization during model training. It also highlights the need to focus on evaluation metrics like recall, F1-score, and the Area Under the Receiver Operating Characteristic Curve (ROC-AUC) [10].

After checking the structure, a linear correlation analysis was performed on the numerical variables. This analysis included the newly created features: login_attempt_ratio and amount_account_age_ratio.

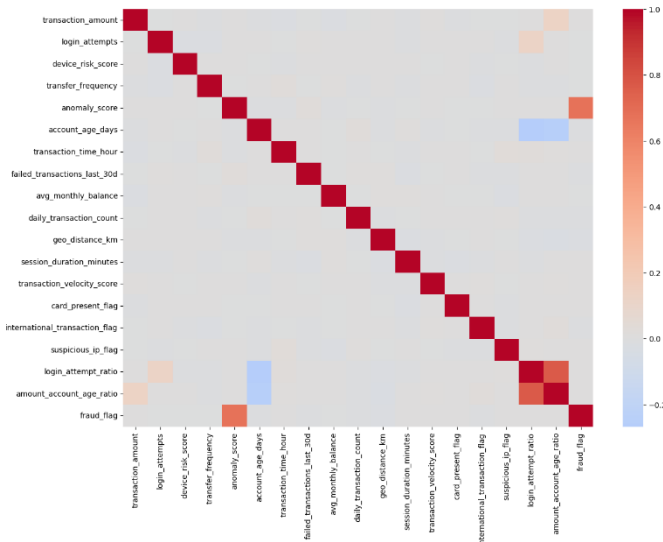


Figure 3. Correlation Heatmap of Numerical Features

The correlation matrix heatmap in Figure 3 shows several key behavioral insights. The anomaly_score variable has the strongest positive linear correlation with the fraud_flag target. This relationship suggests that statistical deviations from normal customer behavior may act as an important risk indicator. The two engineered metrics show a strong positive correlation with each other. They also maintain an inverse linear relationship with account age days. This pattern confirms that the custom features work as expected by increasing the mathematical importance of login frequency and financial spending on new bank accounts.

A broader examination of the other feature blocks in Figure 2 shows that most raw transactional variables have very weak linear correlations with the target class. This general lack of strong linear relationships means that fraudulent activities cannot be easily separated using simple linear divisions or static rule systems. The patterns of retail banking fraud are multi-dimensional and structurally complex. These findings highlight the need to use advanced non-linear classification models, such as tree-based and ensemble frameworks, which can capture deep feature interactions within the data [14].

An outlier assessment was conducted using boxplots across all numerical variables, as shown in Figure 4.

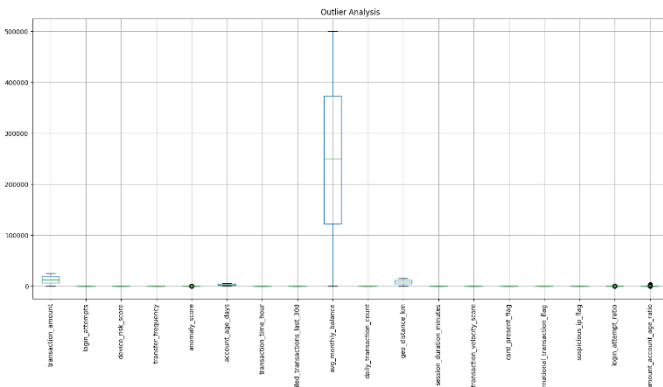


Figure 4. Boxplot Distribution of Numerical Attributes

The analysis revealed a high number of extreme values, especially within continuous financial metrics like avg_monthly_balance and transaction_amount. In standard data cleaning, extreme values are often removed to reduce variance and help the model settle. In fraud analytics, unusual financial patterns and high transaction amounts are indicators of actual fraudulent activities. For this reason, all identified outliers were kept in the training dataset to protect these important risk signals. This ensures that the final models learn from authentic risk patterns [15].

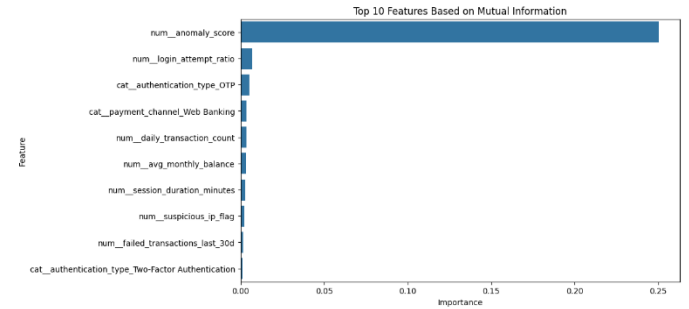


Figure 5. Mutual Information Feature Importance Scores

As shown in Figure 5, the Mutual Information framework calculates the non-linear dependency between each feature and the target variable. A higher score means the feature has a better ability to separate legitimate and fraudulent transactions. Matching the correlation heatmap results, num_anomaly_score is the most critical predictive feature with a score of about 0.25. The engineered feature num_login_attempt_ratio achieved the second-highest score. This result shows that the custom behavior features added valuable information to the model. Categorical security indicators, such as one-time passwords (OTP) and web banking options, also showed clear relationships with fraud. All features were kept for model training to give the classifiers full access to the complete behavioral patterns in the transaction dataset.

B. Model Performance Comparison

The performance of the three machine learning models was evaluated using accuracy, precision, recall, F1-score, and Area Under the Receiver Operating Characteristic Curve (ROC-AUC). The complete comparative metrics are structured in Table II.

TABLE II. COMPARATIVE PERFORMANCE OF CLASSIFICATION MODELS

Method	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.9257	0.6357	0.9493	0.7615	0.9789
Random Forest	0.9433	0.8042	0.7227	0.7612	0.9742
Decision Tree	0.9110	0.5906	0.9387	0.7250	0.9446

Based on the empirical results, Random Forest achieved the highest overall accuracy at 94.33% and the highest precision at

80.42%. This baseline indicates that the Random Forest model was effective in minimizing false fraud predictions. However, in financial risk contexts, classification accuracy is not a reliable metric on its own because failing to catch fraudulent activities causes direct financial damage. For this reason, recall and ROC-AUC are prioritized to evaluate model safety.

Logistic Regression achieved the highest recall value of 94.93%, which shows that the model successfully identified the vast majority of actual fraudulent events. By comparison, Random Forest dropped to a lower recall of 72.27%, meaning it allowed more fraudulent transactions to pass undetected. Logistic Regression also achieved the top ROC-AUC score of 0.9789. This score confirms it has the strongest mathematical capacity to separate fraudulent transactions from legitimate instances across different operational decision boundaries.

The F1-scores for Logistic Regression and Random Forest were highly competitive. Logistic Regression maintained a small advantage with a score of 0.7615 compared to 0.7612 for Random Forest. This margin indicates that Logistic Regression balances precision and recall more effectively given the class distribution of this dataset.

C. ROC Curve Analysis

The Receiver Operating Characteristic (ROC) curve was generated to visually compare the true positive and false positive rates of the three classifiers across all available thresholds.

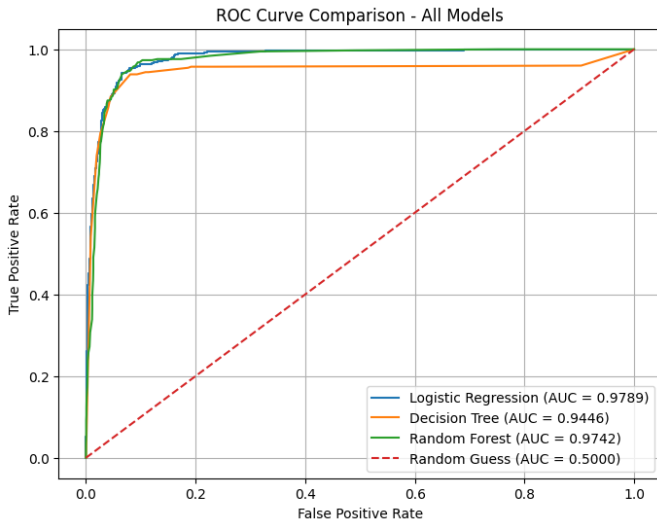


Figure 6. ROC Curve Comparison Across Classifiers

The visual trends presented in Figure 6 confirm the previous numerical evaluations. Logistic Regression generated the largest area under the curve, proving its superior capability to distinguish between fraud and legitimate transactions. This high ROC-AUC performance shows that the Logistic Regression model can rank suspicious transactions effectively. Proper transaction ranking is highly useful for automated banking security platforms that flag high-risk transactions for immediate human review.

D. Model Generalization and Overfitting Analysis

To evaluate model reliability, the difference between training and testing accuracy values was analyzed. The comparative results are presented in Table III.

TABLE III. MODEL GENERALIZATION AND OVERFITTING ANALYSIS

Method	Training Accuracy	Testing Accuracy	Difference
Logistic Regression	0.9190	0.9257	-0.0067
Decision Tree	0.9107	0.9110	-0.0003
Random Forest	1.000	0.9433	0.0567

The results show that Logistic Regression achieved highly similar training and testing performance. This pattern indicates strong generalization ability with a minimal risk of overfitting. The Decision Tree model also showed a small performance gap, but its overall predictive accuracy remained lower compared to the other algorithms. Random Forest achieved a perfect training accuracy of 1.0000 but a lower testing accuracy of 0.9433, creating the largest performance gap in the analysis. This difference indicates that the ensemble model learned the training patterns too closely and displayed clear overfitting behavior. Although Random Forest still achieved high testing accuracy, its ability to generalize to completely unseen transactions was lower than its training performance. Overall, Logistic Regression provided the best balance between predictive power and generalization consistency across both datasets [16].

E. Model Selection

Based on the overall evaluation, Logistic Regression was selected as the best baseline model for the next optimization stage. Although Random Forest achieved higher overall accuracy and precision, Logistic Regression was preferred because banking fraud detection systems must minimize missed fraud cases. The model achieved the top recall value of 94.93%, which shows a stronger capability in identifying hidden fraudulent transactions. Logistic Regression also achieved the highest ROC-AUC score of 0.9789, proving its superior classification ability across multiple decision thresholds. The model showed a much lower risk of overfitting compared to the Random Forest architecture. Logistic Regression was considered the most suitable candidate for final hyperparameter tuning.

F. Hyperparameter Tuning

Hyperparameter tuning was performed on Logistic Regression using the GridSearchCV method with 5-fold cross-validation. This tuning process aimed to find the optimal model parameters to maximize classification performance. The specific parameters evaluated were the regularization strength (C), the regularization type (penalty), and the optimization algorithm (solver). The best parameter combination obtained was C=0.01, penalty='l2', and solver='lbfgs'. The best cross-validation ROC-AUC score achieved during this tuning process was 0.9759. After applying this optimized model configuration to the testing dataset, the final performance results were recorded in Table IV.

TABLE IV. PERFORMANCE METRICS OF TUNED LOGISTIC REGRESSION

Metric	Precision	Recall	F1-Score	ROC-AUC
Score	0.63	0.96	0.76	0.9797



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The tuned Logistic Regression model maintained a strong fraud detection capability, achieving an improved recall value of 0.96. This result indicates that the optimized configuration successfully detected 96% of the actual fraudulent transactions in the test set. The ROC-AUC value also increased slightly from 0.9789 to 0.9797, demonstrating a small improvement in the model's ability to distinguish between fraudulent and legitimate transactions.

V. CONCLUSION

This study developed a machine learning fraud detection framework for retail banking transactions by comparing three classification algorithms: Logistic Regression, Decision Tree, and Random Forest. The experimental dataset contained 10,000 transaction records with a moderate class imbalance, where fraudulent transactions represented exactly 12.5% of the total volume.

Several data preprocessing techniques were executed, including duplicate removal, feature engineering, feature scaling, categorical encoding, and Mutual Information-based feature importance analysis. Two behavioral features were engineered to capture suspicious transaction patterns: the login attempt ratio and the transaction amount per account age ratio.

The experimental results showed that Random Forest achieved the highest overall accuracy and precision. However, Logistic Regression demonstrated superior fraud detection capabilities by achieving the top recall and ROC-AUC scores. Logistic Regression also showed better generalization consistency with a lower risk of overfitting compared to the Random Forest model.

After hyperparameter tuning, the optimized Logistic Regression model achieved a testing ROC-AUC score of 0.9797 and maintained a high recall value of 96%. This performance profile makes it a highly effective model for banking fraud detection applications where missing fraudulent transactions must be avoided.

Future work could explore testing other ensemble methods, applying advanced class imbalance handling techniques, and evaluating different gradient boosting algorithms to improve predictive performance.

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